

Phonon — R^2 heatmap ($n_{\text{train}} = 500$, best PCA per surrogate)

GP

MTGP

DGP

Property

 ϵ_∞ ϵ_0 PhDOS
peak

-0.00

-0.00

-0.00

-0.00

-0.00

-0.00

-0.00

-0.00

0.81

0.91

0.71

0.82

MACE

ORB

SOAP

UMA

Featurizer

-0.00

-0.00

-0.00

-0.00

-0.00

-0.00

-0.00

-0.00

0.80

0.91

0.72

0.80

MACE

ORB

SOAP

UMA

Featurizer

-0.00

-0.00

-0.00

-0.00

-0.00

-0.00

-0.00

-0.00

0.60

0.91

0.64

0.78

MACE

ORB

SOAP

UMA

Featurizer

 R^2

1.0

0.8

0.6

0.4

0.2

0.0